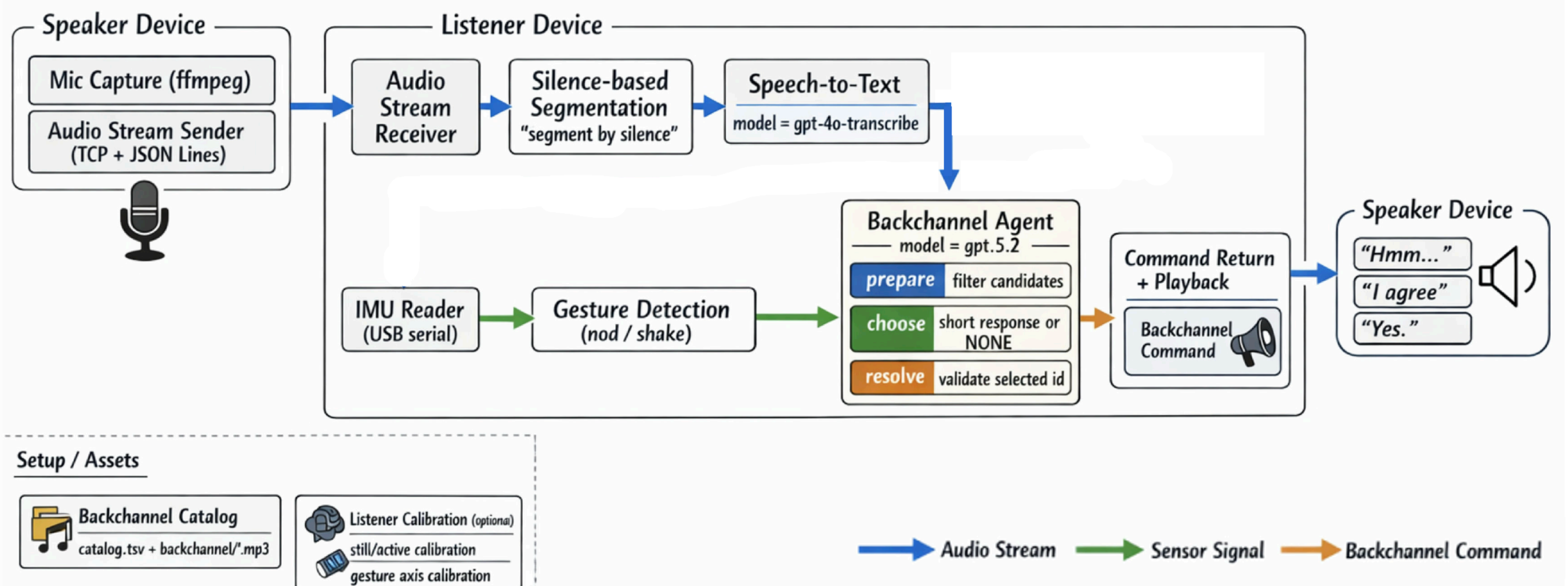
Speaker Device
Mic Capture (ffmpeg)
Audio Stream Sender
(TCP + JSON Lines)
Listener Device
Audio Stream Receiver
Silence-based Segmentation
"segment by silence"
Speech-to-Text
model = gpt-4o-transcribe
IMU Reader
(USB serial)
Gesture Detection
(nod / shake)
Backchannel Agent
model = gpt.5.2
prepare
filter candidates
choose
short response or NONE
resolve
validate selected id
Command Return + Playback
Backchannel Command
Speaker Device
"Hmm..."
"I agree"
"Yes."
Setup / Assets
Backchannel Catalog
catalog.tsv + backchannel/*.mp3
Listener Calibration (optional)
still/active calibration
gesture axis calibration
Audio Stream
Sensor Signal
Backchannel Command